LLMs Are Superhuman Forecasters

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Abstract

In this technical report, we describe a superhuman AI forecasting system. The system is designed to provide calibrated probabilities for a wide range of queries, from political events to global conflicts. Unlike traditional human forecasters or prediction markets, the forecasting AI rapidly aggregates and analyzes news and opinion sources, generating forecasts that are both faster and more cost-effective. Our evaluations on historical questions from the Metaculus forecasting platform demonstrate that the system matches the accuracy of expert human crowds. The demo is available at https://forecast.safe.ai.

1 Background

Policymakers at the highest echelons of government and corporate power have difficulty making high-quality decisions on complicated topics. As the world grows increasingly complex, even coming to a consensus agreement on basic facts is becoming more challenging, as it can be hard to absorb all the relevant information or know which sources to trust. Separately, online discourse could be greatly improved. Discussions on uncertain, contentious issues all too often devolve into battles between interest groups, each intent on name-calling and spouting the most extreme versions of their views through highly biased op-eds and tweets.

Forecasting [1] could provide a solution. Hendrycks et al. [2] proposed the idea of using AI systems as forecasters. In subsequent work, Zou et al. [3] concretized this idea and built a complex AI forecasting system, but it was below crowd level. Likewise, Halawi et al. [4] introduced an AI system with stronger performance, though it had a complicated design. In this brief technical report, we show that a well-engineered prompt for modern LLMs is all that is necessary for attaining roughly crowd-level (superhuman) performance. A demo is available at https://forecast.safe.ai and the code for the demo is available at https://github.com/centerforaisafety/forecasting.

2 Method

The method's retrieval prompt is Figure 1, and the main processing prompt is Figure 2. These prompts are evolved forms of some of the prompts from Halawi et al. [4]. The method is given a question, and associated queries are generated by an underlying LLM following the retrieval prompt. Then after the articles are retrieved, they are pasted into the main processing prompt. The model then follows the steps of the processing prompt, and a final probabilistic prediction is issued.

3 Experiments

Performance. To evaluate the performance of the model, we perform retrodiction, pioneered in Zou et al. [3]. That is to say, we take questions about past events that resolve after the model's pretraining data cutoff date. We then compare the accuracy of the crowd with the accuracy of the model, both

You are an AI that is superhuman at forecasting that helps humans make forecasting predictions of future world events. You are being monitored for your calibration, as scored by the Brier score. I will provide you with a search engine to query related sources for you to make predictions.

First, write breadth google search queries to search online that form objective information for the following forecasting question: {question}

RULES:
0. Your knowledge cutoff is October 2023. The current date is {today}.
1. Please only return a list of search engine queries. No yapping! No description of the queries!
2. Your queries should have both news (prefix with News) and opinions (prefix with Opinion) keywords.
3. Return the search engine queries in a numbered list starting from 1.

Figure 1: Retrieval prompt for our system. Note that it instructs the models to get news and opinion articles.

having access to the same amount of recent information. When we retrieve articles for the forecasting AI, we use the search engine's date cutoff feature, so as not to leak the answer to the model.

We collect 177 Metaculus questions for retrodiction. For each question, we sample 5 forecasts. All metrics are averaged across these forecasts. The accuracy of the Metaculus crowd is 87.0%. Our forecasting system gets 87.7%. If we swap out our prompt with Halawi et al. [4]'s best-performing prompt, we get an accuracy of 83.3%.

Platt scaling. For a set of forecasting questions and labels $\{x_k, y_k\}_{k=1}^K$ such that $y_k \in \{0, 1\}$, we can calibrate the probabilities of the forecasts with Platt scaling. Given an initial forecasted probability $p(x_k)$ for example x_k, the hyperparameter a can be found by minimizing

$$\sum_{k=1}^{K} \left(\sigma \left(a \times \log \frac{p(x_k)}{1 - p(x_k)} \right) - y_k \right)^2.$$

Given this hyperparameter, the new forecasted probability is $\sigma \left(a \times \log \frac{p(x_k)}{1-p(x_k)} \right)$.

On the Metaculus dataset, the crowd gets a Brier score of 0.0952. Our forecasting system without Platt scaling gets 0.0999. With Platt scaling, we get a Brier score of 0.0934. The prompt of Halawi et al. [4] obtains a Brier score of 0.1221.

4 Limitations and Conclusion

We described an AI forecasting system with superhuman forecasting ability. On Metaculus questions, the forecasting AI matches the accuracy of crowd forecasts. Since crowd forecasts are superhuman, so is the forecasting AI. However, many limitations and opportunities for improvement remain. The bot is not fine-tuned, and doing so could potentially make it far more accurate. Forecasts are approximate and do not necessarily sum to 1. Consistency checks of various kinds could address this issue. Finally, a reject option could enable the system to appropriately handle a wider range of questions.

References

[1] Philip E Tetlock and Dan Gardner. Superforecasting: The art and science of prediction. Random House, 2016.

[2] Dan Hendrycks, Nicholas Carlini, John Schulman, and Jacob Steinhardt. Unsolved problems in ml safety. arXiv, 2021.

[3] Andy Zou, Tristan Xiao, Ryan Jia, Joe Kwon, Mantas Mazeika, Richard Li, Dawn Song, Jacob Steinhardt, Owain Evans, and Dan Hendrycks. Forecasting future world events with neural networks. arXiv preprint arXiv:2206.15474, 2022.

[4] Danny Halawi, Fred Zhang, Chen Yueh-Han, and Jacob Steinhardt. Approaching human-level forecasting with language models. ArXiv, abs/2402.18563, 2024.

[5] Nassim Nicholas Taleb and Philip E. Tetlock. On the difference between binary prediction and true exposure with implications for forecasting tournaments and decision making research. 2013.

[6] Philip E. Tetlock, Yunzi Lu, and Barbara A. Mellers. False dichotomy alert: Improving subjective-probability estimates vs. raising awareness of systemic risk. International Journal of Forecasting, 2022.

[7] Sven Ove Hansson. Fallacies of risk. Journal of Risk Research, 2004.

[8] Nassim Nicholas Taleb, Yaneer Bar-Yam, and Pasquale Cirillo. On single point forecasts for fat-tailed variables. International Journal of Forecasting, 2020.

[9] Kailas Vodrahalli, Tobias Gerstenberg, and James Zou. Uncalibrated models can improve human-ai collaboration. ArXiv, abs/2202.05983, 2022.

[10] James Hedlund. Risky business: safety regulations, risk compensation, and individual behavior. Injury Prevention, 2000.

A Limitations of Forecasting

We will now repeat various limitations of relying on AIs for forecasting, quoted almost verbatim from the appendix of Zou et al. [3].

1. Currently, forecasting is best used for refining understanding rather than for anticipating the future more generally. Forecasters are demonstrated to be useful for optimizing probabilities for somewhat likely events (e.g., events with probabilities between, say, 5% and 95%). What is more important are tools that unearth important considerations that were implicitly assigned negligible probabilities or wrongly treated by humans as misinformation or worth ignoring. These considerations are often not forecasted and are not thought worth asking; implicitly, such events could the thought to be assigned low probabilities (e.g., say 10^{-7}), while some people argue that these considerations are more likely than others believe (e.g., say 10^{-1}). The information value provided from putting ignored considerations on our radar is substantial, in fact, orders of magnitude greater than the information gained by refining probabilities by a few percent. Forecasting competitions are about refining estimates of known unknowns—questions already on our radar—but what is better for risk reduction is confronting unknown unknowns, finding considerations to put on our radar, and reducing *exposure* to inchoate hazards. For this reason, Hendrycks et al. [2] also suggest tools that improve brainstorming and suggesting considerations.
2. Forecasting is not necessarily a suitable tool for addressing tail risks. [5] remind us that "No one has yet figured out how to design a forecasting tournament to assess the accuracy of probability judgments that range between .00000001% and 1%—and if someone ever did, it is unlikely that anyone would have the patience–or lifespan–to run the forecasting tournament for the necessary stretches of time (requiring us to think not just in terms of decades, centuries and millennia)." [5] further remind us that it is unjustified to use forecasting tools for revolutions, market crashes, venture capital, or other winner-take-all domains. Furthermore they note that framing questions about tail risks as "a binary question is dangerous because it masks exponentially escalating tail risks." Consequently, "improving short-run probability judgments" and "contingency planning for systemic [tail] risks" are "complementary" and separate [6]. Indeed, superforecasters usually anchor in outside view [1], which neglects systemic risks. In environments with tail events, it is not how often one is correct that matters but rather how large one's cumulative errors are; current forecasting metrics do not sufficiently penalize forecasters that ignore tail risks nor do they greatly reward prescience about Black Swans.
3. Forecasting tools could lead to risky behavior. For example, forecasting systems may induce inaction. If forecasts are uncertain, leaders may argue that "we should not make a decision before we have a reliable forecast" so we should "sit tight and assess." This is sometimes referred to as the delay fallacy, namely "if we wait we will know more about X, hence no decision about X should be made now" [7]. However, it is often cheaper to prevent risks or reduce exposure to risks, as "an existential risk needs to be killed in the egg, when it is still cheap to do so" [8]. Waiting until all the relevant information arrives is often waiting until it is too late.

   Furthermore, humans are known to misinterpret probabilities [9]. Systems that assign an event 3% probability may lead decision-makers to assume the event will not happen. Automation bias may mean forecasting systems induce users to have a gain in confidence that is greater than their gain in knowledge. Risk compensation suggests this could result in riskier actions [10]. Furthermore, forecasts are often not provided with reverse psychology in mind. However, a forecasting system that forecasts a low risk can lead users to act as though there is no risk and increase risky behavior, which increases systemic risk.

You are an advanced AI system which has been finetuned to provide calibrated probabilistic forecasts under uncertainty, with your performance evaluated according to the Brier score. When forecasting, do not treat 0.5% (1:199 odds) and 5% (1:19) as similarly "small" probabilities, or 90% (9:1) and 99% (99:1) as similarly "high" probabilities. As the odds show, they are markedly different, so output your probabilities accordingly.

Question:
{question}

Today's date: {today}
Your pretraining knowledge cutoff: October 2023

We have retrieved the following information for this question:
<background>{sources}</background>

Recall the question you are forecasting:
{question}

Instructions:
1. Compress key factual information from the sources, as well as useful background information which may not be in the sources, into a list of core factual points to reference. Aim for information which is specific, relevant, and covers the core considerations you'll use to make your forecast. For this step, do not draw any conclusions about how a fact will influence your answer or forecast. Place this section of your response in <facts></facts> tags.

2. Provide a few reasons why the answer might be no. Rate the strength of each reason on a scale of 1-10. Use <no></no> tags.

3. Provide a few reasons why the answer might be yes. Rate the strength of each reason on a scale of 1-10. Use <yes></yes> tags.

4. Aggregate your considerations. Do not summarize or repeat previous points; instead, investigate how the competing factors and mechanisms interact and weigh against each other. Factorize your thinking across (exhaustive, mutually exclusive) cases if and only if it would be beneficial to your reasoning. We have detected that you overestimate world conflict, drama, violence, and crises due to news' negativity bias, which doesn't necessarily represent overall trends or base rates. Similarly, we also have detected you overestimate dramatic, shocking, or emotionally charged news due to news' sensationalism bias. Therefore adjust for news' negativity bias and sensationalism bias by considering reasons to why your provided sources might be biased or exaggerated. Think like a superforecaster. Use <thinking></thinking> tags for this section of your response.

5. Output an initial probability (prediction) as a single number between 0 and 1 given steps 1-4. Use <tentative></tentative> tags.

6. Reflect on your answer, performing sanity checks and mentioning any additional knowledge or background information which may be relevant. Check for over/underconfidence, improper treatment of conjunctive or disjunctive conditions (only if applicable), and other forecasting biases when reviewing your reasoning. Consider priors/base rates, and the extent to which case-specific information justifies the deviation between your tentative forecast and the prior. Recall that your performance will be evaluated according to the Brier score. Be precise with tail probabilities. Leverage your intuitions, but never change your forecast for the sake of modesty or balance alone. Finally, aggregate all of your previous reasoning and highlight key factors that inform your final forecast. Use <thinking></thinking> tags for this portion of your response.

7. Output your final prediction (a number between 0 and 1 with an asterisk at the beginning and end of the decimal) in <answer></answer> tags.

Figure 2: Processing prompt for our system.